Expt. No. _____ Page No. _____

Assignment - 6.

② (a) $S_z|\uparrow\rangle = \frac{\hbar}{2}|\uparrow\rangle$; $S_z|\downarrow\rangle = -\frac{\hbar}{2}|\downarrow\rangle$.

$\xrightarrow{\psi_0^0} H|\uparrow\rangle = \frac{\omega\hbar}{2}|\uparrow\rangle$

$H|\downarrow\rangle = -\frac{\hbar\omega}{2}|\downarrow\rangle$

$\therefore E_0 = -\frac{\hbar\omega}{2}$; $E_1 = \frac{\hbar\omega}{2}$ [Ans]

$\therefore |\psi_0^0\rangle = |\downarrow\rangle$; $|\psi_1^0\rangle = |\uparrow\rangle$.

(b) Ground state : $(|\downarrow\rangle)$

$\therefore E_0^1 = \langle\downarrow|H_P|\downarrow\rangle = g\langle\downarrow|S_x|\downarrow\rangle$.

$S_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$\therefore S_x|\downarrow\rangle = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\hbar}{2}|\uparrow\rangle$.

$\therefore S_x|\uparrow\rangle = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{\hbar}{2}|\downarrow\rangle$.

$\therefore E_0^1 = g\frac{\hbar}{2}\langle\downarrow|\uparrow\rangle = 0$

$\therefore E_0^2 = \sum_{k\neq 0} \frac{|\langle k|gS_x|0\rangle|^2}{E_0^0 - E_k^0} = g^2 \frac{|\langle\uparrow|S_x|\downarrow\rangle|^2}{-\frac{\hbar\omega}{2} - \frac{\hbar\omega}{2}}$ [k=1].

$= -g^2\frac{\hbar^2}{4\hbar\omega} = -\frac{g^2\hbar}{4\omega}$.

1st excited state :

Teacher's Signature _____

$E_1^1 = \langle\uparrow|gS_x|\uparrow\rangle = g\langle\uparrow|S_x|\uparrow\rangle = 0.$

$$E_1^2 = \sum_{k\neq 1} \frac{|\langle k|gS_x|1\rangle|^2}{E_1^0 - E_k^0} = g^2 \frac{|\langle\downarrow|S_x|\uparrow\rangle|^2}{\frac{\hbar\omega}{2} + \frac{\hbar\omega}{2}} \quad [k=0].$$

$$= g^2\frac{\hbar^2}{4}\cdot\frac{1}{\hbar\omega} = \frac{g^2\hbar}{4\omega}.$$

$\therefore E_0^{(1)} = 0 = E_1^1$

$E_0^{(2)} = -\frac{g^2\hbar}{4\omega}$; $E_1^2 = \frac{g^2\hbar}{4\omega}$ [Ans]

$$|\psi_0^1\rangle = |\downarrow^1\rangle = \frac{\langle\uparrow|gS_x|\downarrow\rangle}{-\frac{\hbar\omega}{2} - \frac{\hbar\omega}{2}}\cdot|\uparrow\rangle.$$

$$= -\frac{g}{\hbar\omega}\cdot\frac{\hbar}{2}\,|\uparrow\rangle = -\frac{g\hbar}{2\omega}|\uparrow\rangle.$$

[Ans]

$$\therefore |\psi_1^1\rangle = |\uparrow^1\rangle = \frac{\langle\downarrow|gS_x|\uparrow\rangle}{\frac{\hbar\omega}{2} + \frac{\hbar\omega}{2}}|\downarrow\rangle = \frac{g}{2\omega}|\downarrow\rangle.$$

$\therefore |\downarrow\rangle_{\text{corrected}} = |\downarrow\rangle - \frac{g}{2\omega}|\uparrow\rangle.$

$|\uparrow\rangle_{\text{corrected}} = |\uparrow\rangle + \frac{g}{2\omega}|\downarrow\rangle.$ [Ans] [valid if $\frac{g}{\omega} \ll 1$].

(c) $H_t = H + H_p = \omega S_z + gS_x.$

$$H_t = \frac{\hbar}{2}\begin{pmatrix} \omega & g \\ g & -\omega \end{pmatrix}$$

$\therefore$ let its eigenvalues be c

$$\therefore |H_t - cI| = 0 \Rightarrow \begin{vmatrix} \frac{\hbar\omega}{2} - c & \frac{\hbar g}{2} \\ \frac{\hbar g}{2} & -\frac{\hbar\omega}{2} - c \end{vmatrix} = 0$$

$$\Rightarrow -\left(\frac{\hbar\omega}{2}\right)^2 + c^2 - \left(\frac{\hbar g}{2}\right)^2 = 0.$$

$$\Rightarrow c^2 = \frac{\hbar^2}{4}(\omega^2 + g^2).$$

$$\Rightarrow c = \pm\frac{\hbar}{2}\sqrt{\omega^2 + g^2}.$$

Expt. No. _____ Page No. _____

$$\therefore c = \frac{\hbar}{2}\sqrt{\omega^2+g^2} \approx \frac{\hbar\omega}{2}\left(1+\frac{g^2}{2\omega^2}\right) = \frac{\hbar}{2}\left(\omega + \frac{g^2}{2\omega}\right)$$

Let $|\psi_+\rangle = \begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix}$ be the eigenstate

$$\therefore \frac{\hbar}{2}\begin{pmatrix} \frac{\hbar\omega}{2} - \frac{\hbar}{2}\sqrt{\omega^2+g^2} & g \\ g & -\omega - \sqrt{\omega^2+g^2} \end{pmatrix}\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix} = 0$$

$(\omega + g^2/2\omega)$ $\quad$ $-g^2/2\omega$ $\quad$ $(\omega + g^2/2\omega)$

$$\Rightarrow \left(\omega - \sqrt{\omega^2+g^2}\right)\psi_1 + g\psi_2 = 0$$

$$\Rightarrow \boxed{\frac{\psi_2}{\psi_1} = \frac{g}{2\omega}}$$

$$\Rightarrow \frac{\psi_1}{\psi_2} = \frac{g}{\sqrt{\omega^2+g^2}-\omega}$$

$$\therefore |\psi_+\rangle = \begin{pmatrix}1\\ \frac{g}{2\omega}\sqrt{\frac{\omega^2}{g^2}+1} - \frac{\omega}{g}\end{pmatrix} \approx \begin{pmatrix}1\\ 1-\frac{\omega}{g}+\frac{\omega^2}{2g^2}\end{pmatrix} \approx \begin{pmatrix}1\\ -1\end{pmatrix}$$

$$\therefore c = -\frac{\hbar}{2}\sqrt{\omega^2+g^2} \approx -\frac{\hbar\omega}{2}\left(1+\frac{g^2}{2\omega^2}\right) = -\frac{\hbar}{2}\left(\omega+\frac{g^2}{2\omega}\right)$$

Let $|\psi_-\rangle = \begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix}$.

$$\therefore \frac{\hbar}{2}\begin{pmatrix} \omega + \sqrt{\omega^2+g^2} & g \\ g & -\omega + \sqrt{\omega^2+g^2} \end{pmatrix}\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix} = 0$$

$2\omega + g^2/2\omega$ $\quad$ $-\omega + (\omega + g^2/2\omega)$

$$\Rightarrow +\frac{g^2}{2\omega}\left(\omega + \sqrt{\omega^2+g^2}\right)\psi_2 + g\psi_1 = 0$$

$$\Rightarrow \frac{\psi_2}{\psi_1} = \frac{-\omega - \sqrt{\frac{\omega^2}{g^2}+1}}{\frac{g}{2\omega}} = -\frac{2\omega}{g}$$

$$\therefore |\psi_-\rangle = \begin{pmatrix}1\\ -\frac{\omega}{g} - \sqrt{\frac{\omega^2}{g^2}+1}\end{pmatrix} \approx \begin{pmatrix}1\\ \frac{g}{2\omega}\end{pmatrix} = \begin{pmatrix}-g/2\omega\\ 1\end{pmatrix}$$

Teacher's Signature _____

$$\therefore \; E_0 = -\frac{\hbar\omega}{2}\left(1+\frac{g^2}{2\omega^2}\right) = -\frac{\hbar\omega}{2} - \frac{g^2\hbar}{4\omega}.$$

$$E_1 = \frac{\hbar\omega}{2}\left(1+\frac{g^2}{2\omega^2}\right) = \frac{\hbar\omega}{2} + \frac{g^2\hbar}{4\omega}.$$

From our previous calculations,

$$E_0' = E_0^0 + E_0^1 + E_0^2$$
$$= -\frac{\hbar\omega}{2} - \frac{g^2\hbar}{4\omega}.$$

$$E_1' = E_1^0 + E_1^1 + E_1^2 = \frac{\hbar\omega}{2} + \frac{g^2\hbar}{4\omega}.$$

Hence, the actual energies match perfectly with the energies derived from perturbation theory as long as $\left|\frac{g}{\omega}\right| \ll 1$.

$$\therefore \; |\psi\rangle_{\text{corrected}} = |\psi_+\rangle = \begin{pmatrix} 1 \\ \frac{g}{2\omega} \end{pmatrix}$$

$$|\psi\rangle_{\text{corrected}} = |\psi_-\rangle = \begin{pmatrix} -g/2\omega \\ 1 \end{pmatrix}$$

The correctn in wavefns from perturbatn theory are also in accordance with the actual wavefns.

① (a) For number states $|n\rangle$:

$$a|n\rangle = \sqrt{n}\,|n-1\rangle \quad \text{— (i)} \quad ; \quad a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle \quad \text{— (ii)}$$

$$\therefore \langle n|a|n\rangle = 0 \quad \text{— (iii)} \quad ; \quad \langle n|a^\dagger|n\rangle = 0 \quad \text{— (iv)}$$

$$[\langle n|n\pm 1\rangle = 0]$$

$$\langle n|A|n\rangle = \langle A\rangle_n = \frac{1}{\sqrt{2}}\left(\langle n|a|n\rangle + \langle n|a^\dagger|n\rangle\right)$$

$$\Rightarrow \boxed{\langle A\rangle_n = 0} \quad \text{— (a)} \quad [\text{from (iii) \& (iv)}]$$

(Proved)

$$\langle n|B|n\rangle = \langle B\rangle_n = \frac{1}{\sqrt{2}i}\left(\langle n|a|n\rangle - \langle n|a^\dagger|n\rangle\right) = \boxed{0 = \langle B\rangle_n} \quad \text{— (b)}$$

For coherent states:

$$\text{(v)} \text{ — } a|\alpha\rangle = \alpha|\alpha\rangle \quad ; \quad \langle\alpha|a^\dagger = \langle\alpha|\alpha^* \quad \text{— (vi)}$$

$$\therefore \langle\alpha|A|\alpha\rangle = \frac{1}{\sqrt{2}}\left(\langle\alpha|a|\alpha\rangle + \langle\alpha|a^\dagger|\alpha\rangle\right)$$

$$\text{(e)} \leftarrow = \boxed{\frac{1}{\sqrt{2}}(\alpha + \alpha^*)} = \frac{1}{\sqrt{2}}\cdot 2\,\mathrm{Re}(\alpha)$$

$$\Rightarrow \boxed{\langle A\rangle_\alpha = \sqrt{2}\,\mathrm{Re}(\alpha)} \quad \text{— (e)} \quad \text{(Proved)} \quad [\text{from (v) \& (vi)}]$$

$$\langle\alpha|B|\alpha\rangle = \frac{1}{\sqrt{2}i}\left(\langle\alpha|a|\alpha\rangle - \langle\alpha|a^\dagger|\alpha\rangle\right)$$

$$\text{(f)} \leftarrow = \boxed{\frac{1}{\sqrt{2}i}(\alpha - \alpha^*)} = \frac{1}{\sqrt{2}}\cdot 2i\,\mathrm{Im}(\alpha)$$

$$\Rightarrow \boxed{\langle B\rangle_\alpha = \sqrt{2}\,\mathrm{Im}(\alpha)} \quad \text{— (d)} \quad \text{(Proved)} \quad [\text{from (v) \& (vi)}]$$

(b) For ~~number~~ states:

$$A^2 = \frac{1}{2}(a + a^\dagger)(a + a^\dagger)$$

$$\boxed{A^2 = \frac{1}{2}\left[a^2 + aa^\dagger + a^\dagger a + (a^\dagger)^2\right]}$$

Number states

$$\therefore \langle n|a^2|n\rangle = \langle n|(a^\dagger)^2|n\rangle = 0 \quad \text{— (vii)}$$

$$\therefore \langle n|aa^\dagger|n\rangle = \sqrt{n+1}\,\langle n|a|n+1\rangle = (n+1)\langle n|n\rangle \quad \text{— (viii)}$$

$$= n+1 \quad [\text{from (i) \& (ii)}]$$

$$\therefore \langle n|a^\dagger a|n\rangle = \sqrt{n}\,\langle n|a^\dagger|n-1\rangle$$

$$= n \quad [\text{from (i) \& (ii)}] \quad \text{— (ix)}$$

From (vii), (viii) & (iv):

$$\langle A^2\rangle_n = \frac{1}{2}(n+1+n) = \frac{1}{2}(2n+1) \quad \text{— (x)}$$

$$B^2 = -\frac{1}{2}(a-a^\dagger)(a-a^\dagger)$$

$$= -\frac{1}{2}\left[a^2 - aa^\dagger - a^\dagger a + (a^\dagger)^2\right]$$

$$\Rightarrow \boxed{B^2 = \frac{1}{2}\left[aa^\dagger + a^\dagger a - a^2 - (a^\dagger)^2\right]}$$

$$\langle B^2\rangle_n = \frac{1}{2}[n+1+n] = \frac{1}{2}(2n+1) \quad \text{— (xi)}$$

$$\therefore \Delta A_n = \sqrt{\langle A^2\rangle_n - \langle A\rangle_n^2}$$

$$\boxed{\Delta A_n = \frac{1}{\sqrt{2}}\left(\sqrt{2n+1}\right)}$$ [from (a) & (x)]

(Proved)

$$\therefore \Delta B_n = \sqrt{\langle B^2\rangle_n - \langle B\rangle_n^2}$$

$$\boxed{\Delta B_n = \frac{1}{\sqrt{2}}\left(\sqrt{2n+1}\right)}$$ [from (b) & (xi)]

(Proved)

Coherent states

$$\langle\alpha|a^2|\alpha\rangle = \alpha^2 \quad \text{— (xii)}$$

$$\langle\alpha|(a^\dagger)^2|\alpha\rangle = \alpha^{*2} \quad \text{— (xiii)}$$

$$\langle\alpha|aa^\dagger|\alpha\rangle = \langle\alpha|1+a^\dagger a|\alpha\rangle \qquad ([a,a^\dagger]=1)$$

$$= 1 + \langle\alpha|a^\dagger a|\alpha\rangle$$

$$= 1 + \alpha\langle\alpha|a^\dagger|\alpha\rangle$$

$$\langle\alpha|a^\dagger a^\dagger|\alpha\rangle = 1 + |\alpha|^2 \quad \text{— (xiv)}$$

$$\therefore \langle\alpha|a^\dagger a|\alpha\rangle = |\alpha|^2 \quad \text{— (xv)}$$

Using (xii) — (xv):

$$\langle A^2\rangle_\alpha = \frac{1}{2}\left[\alpha^2 + \alpha^{*2} + |\alpha|^2 + 1 + |\alpha|^2\right]$$

$$\Rightarrow \langle A^2\rangle_\alpha = \frac{1}{2}\left[\alpha^2 + \alpha^{*2} + 2|\alpha|^2 + 1\right] \quad \text{— (xvi)}$$

$$\langle B^2\rangle_\alpha = \frac{1}{2}\left[1 + |\alpha|^2 + |\alpha|^2 - \alpha^2 - \alpha^{*2}\right]$$

$$= \frac{1}{2}\left[1 + 2|\alpha|^2 - \alpha^2 - \alpha^{*2}\right] \quad \text{— (xvii)}$$

$$\therefore \Delta A_\alpha = \sqrt{\langle A^2\rangle_\alpha - \langle A\rangle_\alpha^2}$$

$$= \sqrt{\frac{1}{2}[1 + \cancel{2|\alpha|^2} + \cancel{\alpha^2} + \cancel{\alpha^{*2}}] - \frac{1}{2}(\cancel{\alpha^2} + \cancel{\alpha^{*2}} + \cancel{2|\alpha|^2})}$$

$$\boxed{\Delta A_\alpha = \frac{1}{\sqrt{2}}}$$ [from (e) & (XVI)] (Proved)

$$\therefore \Delta B_\alpha = \sqrt{\langle B^2\rangle_\alpha - \langle B\rangle_\alpha^2}$$

$$= \sqrt{\frac{1}{2}[1 + \cancel{2|\alpha|^2} - \cancel{\alpha^2} - \cancel{\alpha^{*2}}] + \frac{1}{2}(\cancel{\alpha^2} + \cancel{\alpha^{*2}} - \cancel{2|\alpha|^2})}$$

$$\boxed{\Delta B_\alpha = \frac{1}{\sqrt{2}}}$$ [from (f) & (XVII)] (Proved)

(c) Let $\hat{K} = +\frac{1}{2}(\gamma a^{+2} - \gamma^* a^2)$.

$$\therefore \hat{S}(\gamma) = e^{-\hat{K}}$$

$$\therefore K^\dagger = +\frac{1}{2}(\gamma^* a^2 - \gamma a^{+2}) = -\hat{K}$$

$$\therefore \hat{S}^\dagger(\gamma) = e^{-K^\dagger} = e^{+K}$$

$$\therefore \boxed{S^\dagger S = e^{+\hat{K}} e^{-\hat{K}} = 1} \longrightarrow S(\gamma) \text{ is unitary}$$

(Proved)

$$S^\dagger(\gamma)\, a\, S(\gamma) = e^{+\hat{K}}\, a\, e^{-\hat{K}}$$

$$= a + \frac{[K, a]}{1!} + \frac{[K, [K, a]]}{2!} + \cdots$$

$$\therefore [K, a] = \frac{1}{2}[\gamma a^{+2} - \gamma^* a^2, a]$$

$$= \frac{1}{2}\left(\gamma [a^{+2}, a] - \gamma^* \underbrace{[a^2, a]}_{\to 0}\right)$$

$$= \frac{1}{2}\left(\gamma\left(a^+ [a^+, a] + [a^+, a] a^+\right)\right)$$

$$= \frac{\gamma}{2}(-a^+ - a^+) = -\gamma a^+ \quad \text{— (XVIII)}$$

$$\therefore [K, [K, a]] = \frac{1}{2}[\gamma a^{+2} - \gamma^* a^2, -\gamma a^+]$$

$$= -\frac{\gamma^*}{2}\left(\gamma \underbrace{[a^{+2}, a^+]}_{\to 0} - \gamma^* [a^2, a^+]\right)$$

$$= \frac{\gamma\gamma^*}{2}\left(a[a, a^+] + [a, a^+] a\right)$$

$$[K, [K, a]] = \frac{\gamma\gamma^*}{2} a \quad \text{— (XIX)}$$

$\therefore$ substituting (XVIII) & (XIX) into equation:

$$S^+ \hat{a} \hat{S} = a + -\frac{\gamma a^+}{1!} + \frac{\gamma\gamma^* a}{2!} + \cdots$$

$$= a\left(1 + \frac{|\gamma|^2}{2!} + \cdots\right) - a^+\left(\frac{\gamma}{1!} + \frac{|\gamma|^2 \gamma}{3!} + \cdots\right)$$

$$= a\left(1 + \frac{r^2}{2!} + \cdots\right) - a^+\left(\frac{r e^{i\theta}}{1!} + \frac{r^3 e^{i\theta}}{3!} + \cdots\right)$$

$$\boxed{S^+ a S = a \cosh(r) - a^+ e^{i\theta} \sinh(r)} \quad \text{— (9)}$$

(Proved)

Similarly:

$$S^+ a^+ S = e^{+K} a^+ e^{K}$$

$$= a^+ + \frac{[K, a^+]}{1!} + \frac{[K, [K, a^+]]}{2!} + \cdots$$

$$[K, a^+] = \frac{1}{2}[\gamma a^{+2} - \gamma^* a^2, a^+]$$

$$= \frac{1}{2}\left(\gamma \underbrace{[a^{+2}, a^+]}_{\to 0} - \gamma^* [a^2, a^+]\right)$$

$$= -\frac{\gamma^*}{2}\left(a[a, a^+] + [a, a^+] a\right) = -\frac{\gamma^* a}{2} \quad \text{— (XX)}$$

$$[K,[K,a^\dagger]] = \frac{1}{2}[\gamma a^{\dagger 2} - \gamma^* a^2, -\gamma^* a]$$

$$= -\frac{\gamma^*}{2}\left(\gamma [a^{\dagger 2}, a] - \gamma^* \cancel{[a^2, a]}^{\to 0}\right)$$

$$= -\frac{\gamma^*\gamma}{2}\left(a^\dagger [a^\dagger, a] + [a^\dagger, a]\, a^\dagger\right)$$

$$[K,[K,a^\dagger]] = |\gamma|^2 a^\dagger \quad \text{— (XXI)}$$

Using (XX) & (XXI):

$$S^\dagger a^\dagger S = a^\dagger + \frac{[K,a^\dagger]}{1!} + \frac{[K,[K,a^\dagger]]}{2!} + \dots$$

$$= a^\dagger + -\frac{\gamma^* a}{1!} + \frac{|\gamma|^2 a^\dagger}{2!} + -\frac{\gamma^* |\gamma|^2 a}{3!} + \dots$$

$$= a^\dagger\left(1 + \frac{r^2}{2!} + \dots\right) - a\left(\frac{\gamma^*}{1!} + \frac{\gamma^* |\gamma|^2}{3!} + \dots\right)$$

$$\Rightarrow \boxed{S^\dagger a^\dagger S = a^\dagger \cosh(r) - a e^{-i\theta} \sinh(r)} \quad \text{(Answered)}$$

— (h)

(d) $\langle A\rangle_\gamma = \langle\gamma|A|\gamma\rangle$

$$= \langle 0|S^\dagger A S|0\rangle$$

$$= \frac{1}{\sqrt{2}}\langle 0|S^\dagger (a + a^\dagger) S|0\rangle$$

[from (g) & (h)]

$$= \frac{1}{\sqrt{2}}\left[\langle 0|S^\dagger a S|0\rangle + \langle 0|S^\dagger a^\dagger S|0\rangle\right]$$

$$= \frac{1}{\sqrt{2}}\left[\langle 0|a\cosh(r) - a^\dagger e^{i\theta}\sinh(r)|0\rangle + \langle 0|a^\dagger\cosh(r) - a e^{-i\theta}\sinh(r)|0\rangle\right]$$

$$\boxed{\langle A\rangle_\gamma = 0} \quad [\langle 0|a|0\rangle = \langle 0|a^\dagger|0\rangle = 0]$$

(Proved)

$$\therefore \langle B\rangle_\gamma = \langle\gamma|B|\gamma\rangle$$

$$= \langle 0|S^\dagger B S|0\rangle$$

[from (g) & (h)]

$$= \frac{1}{\sqrt{2}i}\langle 0|S^\dagger (a - a^\dagger) S|0\rangle$$

$$= \frac{1}{\sqrt{2}i}\left[\langle 0|a\cosh(r) - a^\dagger e^{i\theta}\sinh(r)|0\rangle - \langle 0|a^\dagger\cosh(r) - a e^{-i\theta}\sinh(r)|0\rangle\right]$$

$$\boxed{\langle B\rangle_\gamma = 0} \quad \text{(Proved)}$$

For uncertainty calculations, $\theta = 0$

$$\therefore S^\dagger A S = \frac{1}{\sqrt{2}}\left[S^\dagger (a + a^\dagger) S\right]$$

$$= \frac{1}{\sqrt{2}}\left[a\cosh(r) - a^\dagger e^{i\theta}\sinh(r) + a^\dagger\cosh(r) - a e^{-i\theta}\sinh(r)\right]$$

$$= \frac{1}{\sqrt{2}}\left[a\cosh(r) - a^\dagger \sinh(r) + a^\dagger \cosh(r) - a\sinh(r)\right]$$

$$= \frac{1}{\sqrt{2}}\left[\cosh(r) - \sinh(r)\right](a + a^\dagger)$$

$$\boxed{S^\dagger A S = A e^{-r}} \quad \text{(XXII)} \qquad \left[A = \frac{1}{\sqrt{2}}(a + a^\dagger) \; ; \; \cosh(r) - \sinh(r) = e^{-r}\right]$$

$$\because \langle A\rangle_\gamma = 0 \; ; \quad (\Delta A)^2_\gamma = \langle A^2\rangle_\gamma$$

$$= \langle \gamma | A^2 | \gamma\rangle$$

$$= \langle 0 | S^\dagger A^2 S | 0\rangle$$

$$= \langle 0 | (S^\dagger A S)\cdot(S^\dagger A S) | 0\rangle$$

$$= \langle 0 | A^2 e^{-2r} | 0\rangle \quad [\text{from (XXII)}]$$

$$= e^{-2r}\langle 0 | A^2 | 0\rangle$$

$$\Rightarrow \boxed{(\Delta A_\gamma)^2 = \frac{1}{2}e^{-2r}} \quad [\text{from (X)}] \quad \text{(i)}$$

$$e^{-2r} = [\cosh(r) - \sinh(r)]^2$$

$$= \cosh^2(r) + \sinh^2(r) - 2\cosh(r)\sinh(r)$$

$$\therefore \boxed{(\Delta A_\gamma) = \frac{1}{\sqrt{2}}\sqrt{\cosh^2(r) + \sinh^2(r) - 2\cosh(r)\sinh(r)}}$$

(Proved)

Similarly, for B :

$$S^\dagger B S = \frac{1}{\sqrt{2}}\left[S^\dagger (a - a^\dagger) S\right]$$

$$= \frac{1}{\sqrt{2}}\left[a\cosh(r) - a^\dagger \sinh(r) - a^\dagger \cosh(r) + a\sinh(r)\right]$$

$$= \frac{1}{\sqrt{2}}(a - a^\dagger)\left[\cosh(r) + \sinh(r)\right]$$

$$\boxed{S^\dagger B S = B e^{r}} \quad \text{(XXIII)}$$

$$\therefore (\Delta B_\gamma)^2 = \langle \gamma | B^2 | \gamma\rangle \qquad [\because \langle B\rangle_\gamma = 0]$$

$$= \langle 0 | S^\dagger B^2 S | 0\rangle$$

$$= \langle 0 | (S^\dagger B S)^2 | 0\rangle$$

$$\Rightarrow \boxed{(\Delta B_\gamma)^2 = \langle 0 | B^2 e^{2r} | 0\rangle = \frac{1}{2}e^{2r}} \quad [\text{from (XI)}] \quad \text{(j)}$$

$$e^{2r} = [\cosh(r) + \sinh(r)]^2$$

$$= \cosh^2(r) + \sinh^2(r) + 2\cosh(r)\sinh(r)$$

$$\therefore \boxed{\Delta B_\gamma = \frac{1}{\sqrt{2}}\sqrt{\cosh^2(r) + \sinh^2(r) + 2\cosh(r)\sinh(r)}}$$

(Proved)

From (i) & (j) : $\boxed{(\Delta A)_\gamma (\Delta B)_\gamma = \frac{1}{2}}$ (Proved)